

SPECTROSCOPY & FLAME TESTS

Elementary Science Series • Flame Colors & Atomic Light

GRADE 5 SCIENCE

Name: _____

Class/Sec: _____

Date: _____

1. KEY CONCEPTS & STORY RECAP: HOW LIGHT IS CREATED

**The Magic of Diwali Fireworks!**

Have you ever wondered why fireworks explode in dazzling reds, bright greens, and glowing yellows? It is all because of **metals**! **Spectroscopy** is the study of how light and matter work together. When we heat metal salts in a flame, they produce colorful light.

The 3-Step "Electron Jump & Flash Dance":

- Heat Energy (Excitation):** Flame heat gives tiny electrons inside atoms a boost of energy, causing them to jump up to a higher energy level.
- Falling Back (Relaxation):** The electrons cannot stay up there for long! They drop back down to their home level.
- Light Release (Emission):** As they fall, they release that extra energy as a flash of colorful light!

Lab Safety & Testing Rules:

- Safety Goggles On:** Always protect your eyes in the lab!
- Clean Wire Loop:** Dip a nichrome wire in acid (HCl) to clean off old chemicals so colors don't mix.
- Blue Flame:** Use the hot blue part of the flame so you can see the color clearly.
- Element Fingerprint:** Every element has a unique color, just like every person has unique fingerprints!

 **Fun Fact for 5th Graders:** Astronomers use spectroscopy to find out what stars millions of miles away are made of! By looking at a star's light color, scientists can tell which elements are burning inside it.

2. ELEMENT & FLAME SPECTRUM REFERENCE TABLE

Element	Symbol	Flame Color	Everyday Real-World Examples
Sodium	Na	 Bright Yellow	Table salt (NaCl), yellow street lights, potato chips.
Potassium	K	 Lilac / Pale Violet	Bananas, plant fertilizer, fireworks.
Copper	Cu	 Blue-Green / Emerald	Copper water pots, pennies, electrical wires, brass keys.
Calcium	Ca	 Brick Red	Milk, chalk, eggshells, limestone, festive diyas.
Strontium	Sr	 Deep Crimson Red	Emergency roadside flares, bright red fireworks.
Barium	Ba	 Apple Green	Green fireworks, glow-in-the-dark paints.
Lithium	Li	 Magenta / Hot Pink	Rechargeable phone batteries, electric car batteries.

3. STUDENT PRACTICE EXERCISES

Q1. Multiple Choice Challenge

a) Which element burns with a bright yellow flame when table salt is sprinkled into a burner flame?

[] (A) Copper (Cu)

[] (B) Sodium (Na)

[] (C) Calcium (Ca)

[] (D) Potassium (K)

b) What causes an atom to emit a flash of colored light during a flame test?

[] (A) Electrons absorbing heat and jumping up

[] (B) Excited electrons falling back down to their home level

[] (C) Protons breaking apart inside the nucleus

[] (D) Water evaporating from the metal wire

Q2. Color Matching activity

Match each metal element on the left with its correct flame test color on the right:

1. **Copper (Cu)** → Letter: _____
2. **Calcium (Ca)** → Letter: _____
3. **Potassium (K)** → Letter: _____
4. **Strontium (Sr)** → Letter: _____

Color Options:

- A. Lilac / Pale Violet
- B. Blue-Green / Emerald
- C. Deep Crimson Red
- D. Brick Red

Q3. Complete the Sentence (Fill in the Blanks)

Use the word bank below to fill in the blanks:

[Fingerprint | Jump | Acid | Spectroscopy | Blue]

1. The scientific study of light and matter is called _____.
2. In the flame test, heat energy makes electrons _____ to a higher level.
3. To clean the metal testing loop, scientists dip it in hydrochloric _____.
4. Every element produces a unique color pattern known as an elemental _____.

Q4. Real-World Detective Case

A firework show technician wants to create a grand finale with bright **green** and deep **crimson red** explosions. Which two metal elements should the technician mix into the fireworks?

Q5. True or False?

- | | |
|---|-----------|
| a) All metal elements emit the exact same color when burned. | [] |
| b) Bananas contain potassium, which turns a flame lilac/violet. | [] |
| c) Calcium is found in milk and chalk and produces a brick red flame. | [] |